

S N	Authors (Publishing Year, Publisher)	Research Problem	Dataset	Methodology (Workflow)	Result Metrics	Findings/Outcome/ Contribution	Limitations	Future Direction and Research Extension Idea
01	Khan et al. (2025) (IEEE QPAIN Conference 2025)	Static feature extraction. Feature redundancy remains unresolved. Limited hyperparameter optimization. Overfitting due to weak regularization. High computational cost in high-dimensional tuning.	CBIS-D DSM (Source : Kaggle) Data Size: 10203 (After Augmentation) 3 Classes: Benign-without-callback (0) Benign (1), Malignant (2).	Deep Learning Models (CNN, ResNET50, DenseNet12,...), Ensemble with Traditional and Quantum-Inspired Hyper Parameter Optimization. CBIS-DDSM Data Collection -> Image Labelling-> Preprocessing, Reshaping, Augmentation, and balancing -> Split Dataset -> Train Dataset-> DL Model Training with Traditional and Quantum-Inspired Tuning -> Ensemble DL Models with best Results-> Compare Results	Accuracy: ResNet 50: 89% (Test), 94%(Train), DenseNet 90% (Test), 96%(Train), Ensemble: 92% (Test), 98% (Train)	Quantum-Inspired Hyperparameter Tuning on Ensemble Model can give efficient, robust accuracy with limited data.	Hybrid Tuning approach can be applied.	Hybrid Hyperparameter Tuning Approach can be implemented. Explore real-world, large datasets.
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References:

1. Khan, N. K., & Islam, M. N. (2025, July). Breast Cancer Detection Using Deep Learning and Quantum-Inspired Hyperparameter Optimization. In *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)* (pp. 1-6). IEEE.